

Project Proposal: Reducing Scrap and Waste in the Lamination Stage of Copper Clad Laminate (CCL) Manufacturing

Problem Statement:

In the production of copper clad laminates (CCL), the **lamination stage** is critical for bonding the prepreg material with copper foil. Defects during this stage, such as delamination, uneven bonding, voids, or incorrect thickness, can lead to high scrap rates, which waste materials and increase production costs. This project will focus on identifying the root causes of these defects in the lamination process and applying quality control (QC) tools to reduce scrap and improve bonding quality.

Project Focus:

The focus of this project is on analyzing and optimizing the lamination stage of CCL manufacturing, where the copper foil is bonded to the prepreg. By applying QC tools, the goal is to reduce defects, enhance material bonding, and improve process consistency, thereby reducing scrap and improving the overall efficiency of the lamination process.

Objectives:

- **Identify common defects** in the lamination stage, such as delamination, uneven bonding, or voids.
- **Map the lamination process** to highlight areas prone to defects and inefficiencies.
- **Monitor and control critical quality parameters** during lamination, such as pressure, temperature, and bonding time.
- **Reduce scrap rates** and improve the bonding quality between the copper foil and prepreg.

Quality Control Tools:

1. Histogram:

- Analyze the frequency of defects related to the lamination stage (e.g., delamination, improper bonding, uneven thickness).
- Identify the most common types of defects and their occurrence rates to prioritize improvement areas.

2. Process Flow Diagram:

- Develop a detailed map of the lamination process, from prepreg preparation to final lamination and quality inspection.
- Identify critical points in the process where defects typically occur, such as during heating or pressing.

3. Control Charts:

- Monitor key quality parameters during the lamination stage, including pressure, temperature, and bonding time.
- Use control charts to ensure that the process remains within specification limits, tracking any deviations that could lead to defects.

Key Process Steps in the Lamination Stage:

1. Prepreg and Copper Foil Preparation:

- Key Quality Parameters: Cleanliness of the copper foil, proper alignment of prepreg layers and copper.
- Potential Defects: Misalignment, contamination leading to poor adhesion.

2. Pre-Lamination Heating:

- Key Quality Parameters: Temperature control to partially cure the resin without causing premature bonding.
- Potential Defects: Overheating or underheating, leading to improper resin flow and bonding issues.

3. Pressing and Bonding:

- Key Quality Parameters: Application of uniform pressure and temperature to ensure even bonding of the prepreg and copper foil.
- Potential Defects: Uneven pressure distribution, resulting in delamination, bubbles, or voids.

4. Post-Lamination Cooling:

- Key Quality Parameters: Controlled cooling to prevent thermal stresses and maintain bond integrity.
- Potential Defects: Warping, cracking, or delamination due to improper cooling rates.

5. Final Inspection and Testing:

- Key Quality Parameters: Laminate thickness, adhesion strength, and surface quality.
- Potential Defects: Inconsistent thickness, weak adhesion, surface imperfections.

Outcomes:

- **Reduced Scrap Rates:** By identifying and addressing the key defects in the lamination stage, the project aims to significantly reduce the amount of material scrapped due to bonding issues.
- **Improved Bonding Quality:** With fewer defects, the bonding between the prepreg and copper foil will be stronger and more consistent, enhancing product quality.

- **Process Consistency:** By using control charts to monitor and maintain key process parameters, the lamination stage will be more stable and produce fewer defective products.
- **Cost Savings:** Reducing scrap and improving bonding efficiency will lead to cost savings in raw materials, labor, and energy consumption